$N(N-1)$. This implies that the right-hand side and the left-hand side are proportional. The proportionality coefficient (which is equal to 1) is found by induction by multiplying both sides by $z_N - y_N$ and then setting $z_N = y_N$. $\square$

Now setting $z_i = 1/x_i$ in the lemma, we get

Corollary 5.15.4 (Cauchy identity)**.**

$$R(x,y) = \det\left(\frac{1}{1-x_iy_j}\right) = \sum_{\sigma\in S_N} \frac{(-1)^\sigma}{\prod_{j=1}^N (1-x_jy_{\sigma(j)})}.$$

Corollary 5.15.4 easily implies that the coefficient of $x^{\lambda+\rho}y^{\lambda+\rho}$ is 1. Indeed, if $\sigma \neq 1$ is a permutation in S_N, the coefficient of this monomial in $\frac{1}{\prod_j(1-x_jy_{\sigma(j)})}$ is obviously zero, since the coordinates of $\lambda+\rho$ are strictly decreasing and hence distinct. $\square$

Remark 5.15.5. For partitions λ and μ of n, let us say that $\lambda \preceq \mu$ or $\mu \succeq \lambda$ if $\mu - \lambda$ is a sum of vectors of the form $e_i - e_j$, $i < j$ (called positive roots). This is a partial order, and $\mu \succeq \lambda$ implies $\mu \geq \lambda$. It follows from Theorem 5.15.1 and its proof that

$$\chi_\lambda = \sum_{\mu\succeq\lambda} \widetilde{K}_{\mu\lambda}\chi_{U_\mu},$$

where $(\widetilde{K}_{\lambda\mu})$ is the matrix inverse to the matrix of Kostka numbers $(K_{\lambda\mu})$. This implies that the Kostka numbers $K_{\mu\lambda}$ vanish unless $\mu \succeq \lambda$.

5.16. Problems

In the following problems, we do not make a distinction between Young diagrams and partitions.

Problem 5.16.1. For a Young diagram μ, let $A(\mu)$ be the set of Young diagrams obtained by adding a square to μ, and let $R(\mu)$ be the set of Young diagrams obtained by removing a square from μ.

(a) Show that $\mathrm{Res}^{S_n}_{S_{n-1}} V_\mu = \bigoplus_{\lambda\in R(\mu)} V_\lambda$.

(b) Show that $\mathrm{Ind}^{S_n}_{S_{n-1}} V_\mu = \bigoplus_{\lambda\in A(\mu)} V_\lambda$.